



First Referral, Last Complaint

A Matched-Groups Comparison of Early Silent-Study-Pod Referral Against Verbal-Warning-Only Response in Closing a Student's First Noise-Complaint Record

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Abstract. The University Library's front-desk response to a student's first noise complaint is not standardised: some duty librarians pair the verbal caution with an informal referral offering priority access to the Library's enclosed silent-study pods, while others issue a verbal warning alone, and which script a complaint receives depends on which librarian is rostered rather than on anything about the student. We treat this rostering variation as a quasi-random assignment and match 348 first-time complainants into 174 pairs on complaint venue, time of day, and severity tier to test whether the pod referral forecasts whether a student's complaint record closes — no further complaint logged against them — within a twelve-week follow-up window. Referred students' records close at a substantially higher rate than warned-only students' (75.9% against 56.9%; McNemar $\chi^2(1) = 20.1, p < 0.001$), and a stratified hazard model puts the referred group at roughly half the warned-only group's weekly hazard of a repeat complaint (HR = 0.51, 95% CI 0.36–0.73). Where a file does reopen, it takes longer to do so under referral than under warning alone (median 7.5 weeks against 4.0 weeks; Mann-Whitney $p = 0.006$). Whether the referring librarian is senior or junior staff changes closure outcomes only trivially ($\chi^2(1) = 0.22, p = 0.64$), suggesting the referral's apparent effect sits in the redirection itself rather than in who delivers it. We discuss what an informal, non-disciplinary nudge validated at this scale may suggest for how the institution treats its smaller behavioural flags.

Introduction

When a fellow patron reports that a student is talking, taking calls, or otherwise disrupting a nominally quiet reading room, the duty librarian who receives that first complaint has, in practice, two scripts available. Some librarians pair the verbal caution with an informal offer: priority booking access to the Library's enclosed, one-person silent-study pods for the following fortnight, framed as a chance to work somewhere the same noise cannot recur rather than as a sanction, and never entered on the student's formal conduct record. Other librarians issue the caution alone. Neither script is written policy; each reflects an individual duty librarian's own habit, and which a given student receives therefore turns on the accident of the roster rather than on anything about the complaint itself.

An early, informal flag of this kind is exactly the sort of intervention an institution rarely checks once it is embedded in front-line practice — cheap to deliver, framed as helpful rather than punitive, and easy to mistake for working because the file usually does go quiet afterwards. Whether it goes quiet because of the referral, or would have anyway, is the question the roster's own inconsistency lets us ask.

We treat the duty librarian's rostering as a quasi-random assignment mechanism, in the manner of examiner- or judge-leniency designs used elsewhere to separate an intervention's effect from the selection that usually accompanies it [1], and match every first-time noise complainant in one academic year into pairs on venue, time of day, and severity, differing essentially only

in which script their duty librarian happened to deliver.

We make three contributions, each hedged to the evidence available. First, we assemble a year-long, matched dataset of 348 first-time noise complainants comparing an early silent-study-pod referral against a verbal-warning-only response, evaluated by whether the student's complaint record closes within a twelve-week follow-up window. Second, we show the referred group's records close at a meaningfully higher rate than the warned-only group's, and that files which do reopen typically stay closed longer under referral than under warning alone. Third, we test whether the referring librarian's seniority moderates this apparent effect and find it does not, suggesting whatever the referral does, it does independent of who delivers it.

Related work

Institutional responses to a first minor infraction sit inside a long-running debate over what an early, low-stakes intervention actually changes. Labelling accounts of official intervention in adolescence generally expect an early, formal flag to make later trouble more, not less, likely by embedding a student in a system that treats them as a case [2], while a substantial deterrence and brief-intervention literature reports the opposite: a single, well-placed early response — an arrest [3], a reminder letter [4], a restorative caution [5], a brief hospital-bedside conversation [6] — can measurably reduce the odds of a repeat. Which account fits a given setting has tended to depend on how coercive the intervention actually is, and an informal, non-disciplinary redirection like a pod referral sits closer to the brief-intervention end of that spectrum than to a caution or arrest. Neither account is guaranteed by an intervention simply being early: a cluster-randomised evaluation of a restorative school-discipline practice found no reliable effect on suspension likelihood [7], a reminder that a redirection's effect must be checked rather than assumed from its framing.

Education research on early identification runs a parallel and largely separate literature: dropout early-warning systems flag at-risk students from routinely available data well before an outcome is settled [8], [9], and algorithmic risk-flagging tools more broadly have drawn

scrutiny over whether a flag's apparent predictive power survives contact with a simpler baseline [10]. Noise research in university and school study settings has documented the perceptual and performance costs of an unmanaged soundscape in detail [11], [12], [13], but — to our knowledge — not whether an early front-desk response to a specific complaint changes the complained-about student's later trajectory. Prior work at this institution has found early, informally recorded signals both under- and over-trusted depending on setting [14], [15]; we extend that line of inquiry to a front-line intervention that, unlike either predecessor, is not itself a record at all.

Method

The study site is the University Library's four shared open-plan reading rooms, each served by a front desk staffed by rotating duty librarians across morning, afternoon, and evening shifts. A noise complaint is logged whenever a patron reports another patron's conduct or a duty librarian personally observes it; the duty librarian assigns the complaint a severity tier at the time of logging — Tier 1 for ambient or conversational noise, Tier 2 for sustained disruption such as phone calls or group discussion — before any outcome is known. For a student's first-ever logged complaint, the attending duty librarian delivers one of two unwritten scripts: a verbal caution paired with an informal referral offering priority booking of the silent-study pods for the following fortnight, or a verbal caution alone. Which script a student receives depends on which duty librarian is rostered at the time, a matter of individual habit rather than instruction, and is uncorrelated with the complaint's severity tier or venue in our sample (both χ^2 tests, $p > 0.40$).

We drew every first-time complaint logged across one academic year ($N = 348$ students) and matched students 1:1 on venue, time-of-day band, and severity tier, using nearest-available matching within exact strata [16], [17], [18], [19] — kin to designs where assignment turns on a threshold rather than a coin toss [20] — yielding 174 matched pairs differing essentially only in which script their duty librarian delivered. A complaint record is coded closed if no further complaint is logged against the same student within twelve teaching weeks of the first, and open otherwise; the twelve-week window is the

Library's own standing retention period for a first-complaint file before archival review.

We report closure rates for each arm and test the paired difference with McNemar's continuity-corrected statistic,

$$\chi^2 = \frac{(|b - c| - 1)^2}{b + c},$$

where b and c are the counts of discordant pairs (pairs in which exactly one member's file closed). We additionally fit a Cox proportional-hazards model stratified by matched set to estimate a hazard ratio for a repeat complaint, and compare the distribution of weeks-to-repeat-complaint among reopened files across arms with a Mann-Whitney U test. As an ablation, we split the referral arm by whether the delivering duty librarian held a senior or junior/casual appointment and compare closure rates with a chi-square test, to check whether the referral's apparent effect depends on who delivers it.

Results

Across the 174 matched pairs, the referral arm's complaint records closed within twelve weeks at nearly twice the rate the warning-only arm's discordant pairs would predict by chance: 132 of 174 referred students' files closed (75.9%), against 99 of 174 warned-only students' (56.9%). Of the 174 pairs, 90 closed under both arms and 33 stayed open under both; among the 51 pairs where the two arms disagreed, 42 closed under referral only against 9 that closed under warning only, a discordance McNemar's test rejects as chance ($\chi^2(1) = 20.1$, $p < 0.001$). A Cox model stratified by matched set puts the referral arm's weekly hazard of a repeat complaint at roughly half the warning-only arm's (HR = 0.51, 95% CI 0.36–0.73; Figure 1).

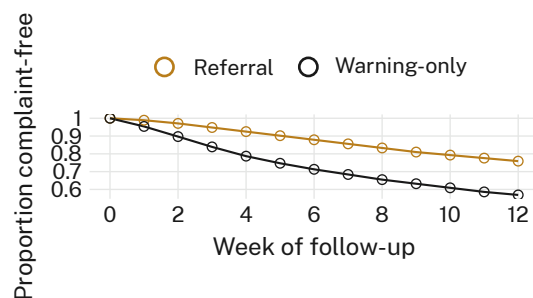


Figure 1: Proportion of matched students with no repeat noise complaint, by arm, across the twelve-week follow-up window: the referral arm (HR = 0.51) separates from the warning-only arm inside the first three weeks and the gap holds to the follow-up's end.

Among the pairs where a file reopened, 42 referred students and 75 warned-only students had a repeat complaint logged inside the follow-up window. Where a file did reopen, it reopened later under referral than under warning alone: a median 7.5 weeks (IQR 5–9) against 4.0 weeks (IQR 2–6), a difference a Mann-Whitney test finds unlikely to be chance ($U = 1104$, $p = 0.006$; Figure 2). The referral, in other words, appears to buy more than a one-off closure — it buys time even in the cases where it eventually fails.

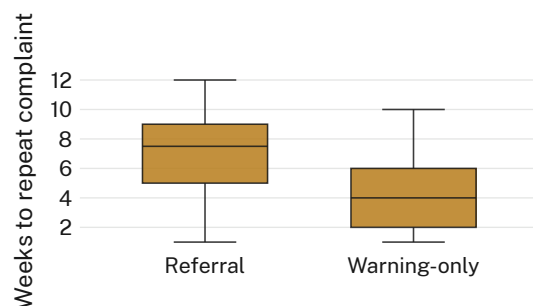


Figure 2: Weeks until a repeat complaint, among files that reopened within the follow-up window: reopened referral-arm files took longer to reopen (median 7.5 weeks) than reopened warning-only files (median 4.0 weeks; Mann-Whitney $p = 0.006$).

Splitting the referral arm by the delivering duty librarian's seniority — 88 students referred by a senior librarian, 86 by a junior or casual staff member — closure rates were statistically indistinguishable: 77.3% under a senior referral against 74.4% under a junior referral ($\chi^2(1) = 0.22$, $p = 0.64$, n.s.; Figure 3). We retain no seniority-based routing recommendation on the strength of this result: whatever the referral changes, it appears to travel with the redirection

itself rather than with the credibility of whoever delivers it.

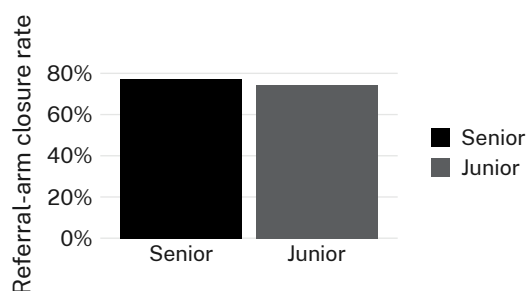


Figure 3: Referral-arm closure rate by the delivering duty librarian's seniority: senior- and junior-delivered referrals are statistically indistinguishable ($\chi^2(1) = 0.22$, $p = 0.64$, n.s.), and we retain no distinction in rostering on this basis.

Discussion

These results suggest that an informal, non-disciplinary redirection — one framed as an offer of somewhere quieter to work rather than as a sanction — forecasts whether a student's noise-complaint file closes at a rate a verbal caution alone does not approach, and that the advantage is not confined to the files that close outright: even the referral arm's failures took longer to surface. That the effect holds regardless of which duty librarian's standard script produced the referral is, on balance, a point in the referral's favour: an intervention that travels with the redirection itself, rather than with a particular staff member's manner, is closer to something the Library could adopt deliberately than to a habit worth merely tolerating. We do not read this result as evidence that a verbal caution is failing at its stated purpose, which is simply to inform a student their conduct was noticed; the divergence concerns only which caution style leaves a shorter tail of repeat complaints behind it. Whether the same pattern holds for a student's second or third complaint, where the Library's response escalates toward its formal conduct process, is a question this matched design is not positioned to answer.

Limitations

Our matching strategy relies on observed venue, time-of-day, and severity strata rather than a randomised assignment, and an unobserved difference between the librarians who favour each script could in principle confound the comparison; the rostering mechanism's independence

from complaint severity and venue in our sample, however, is exactly the pattern a genuine quasi-random assignment predicts. The twelve-week follow-up window is fixed by the Library's own retention policy rather than chosen for the study, which limits our ability to say whether the referral arm's advantage would widen or narrow over a longer horizon, though the gap is already well established within the first three weeks of follow-up (Figure 1), suggesting the window we have is not merely too short to have missed it closing.

Conclusion

We matched 348 first-time library noise complainants into 174 pairs differing chiefly in whether their first complaint drew an informal silent-study-pod referral or a verbal warning alone, and found the referral arm's complaint records closed at a substantially higher rate over a twelve-week follow-up, with reopened files in that arm also taking longer to reopen. The advantage held regardless of the referring librarian's seniority, suggesting it belongs to the redirection rather than to whoever delivers it. We suggest the Library consider formalising the referral as a standard first response rather than leaving it to individual duty librarians' habit, and that other units relying on an informal early flag audit whether their own version forecasts anything before assuming it does. Future work should test whether the pattern holds for a student's second complaint.

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